Subject: Session 3.3
Year. Month. Date. ( )

Problem 27.1

$$ABx = \lambda x$$

$$(AB)^T Bx = \lambda \cancel{B} x^T B x$$

$$(B\cancel{x})^T A^T B x = \lambda x^T B x$$

$$B \cancel{x}^T A^T B x = \lambda x^T \cancel{x} B^T X$$

$$(B\cancel{x})^T A^T B x = \lambda B X$$

$$BA^T > 0 \longrightarrow \lambda > 0$$

Problem 27.2

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 7x_2 & 5x_1 + 9x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= x_1^2 + 7x_1x_2 + 5x_1x_2 + 9x_2^2$$

$$= x_1^2 + 12x_1x_2 + 9x_2^2$$

$$= (x_1 + 3x_2)^2 + 6x_1x_2$$

Sometimes + Sometime −